

IFEval Component-Wise Mixture Fit: $H=4$, $G=(3,3,1,3)$

This note summarizes a focused IFEval fit of the binary probit independent-mixture factor model with

```
H = 4
G = (3, 3, 1, 3)
lambda_l1_penalty = 4
pretraining iterations = 20 maximum
MAP refinement iterations = 20 maximum
```

The fitted full-data model stopped after 15 pretraining iterations and 12 MAP refinement iterations. The full-data binary probit log likelihood was -0.21897 per response. In the completed cross-validation table, this specification has held-out log likelihood -0.26978 per response, very close to the current best component-wise fit $G=(3,3,2,2)$, which has held-out log likelihood -0.26944 per response.

Component-Wise CV Context

The current component-wise CV screen strongly favors $H=4$ and $\lambda=4$. Among these fits, $G=(3,3,1,3)$ is near the top while keeping the third coordinate Gaussian.

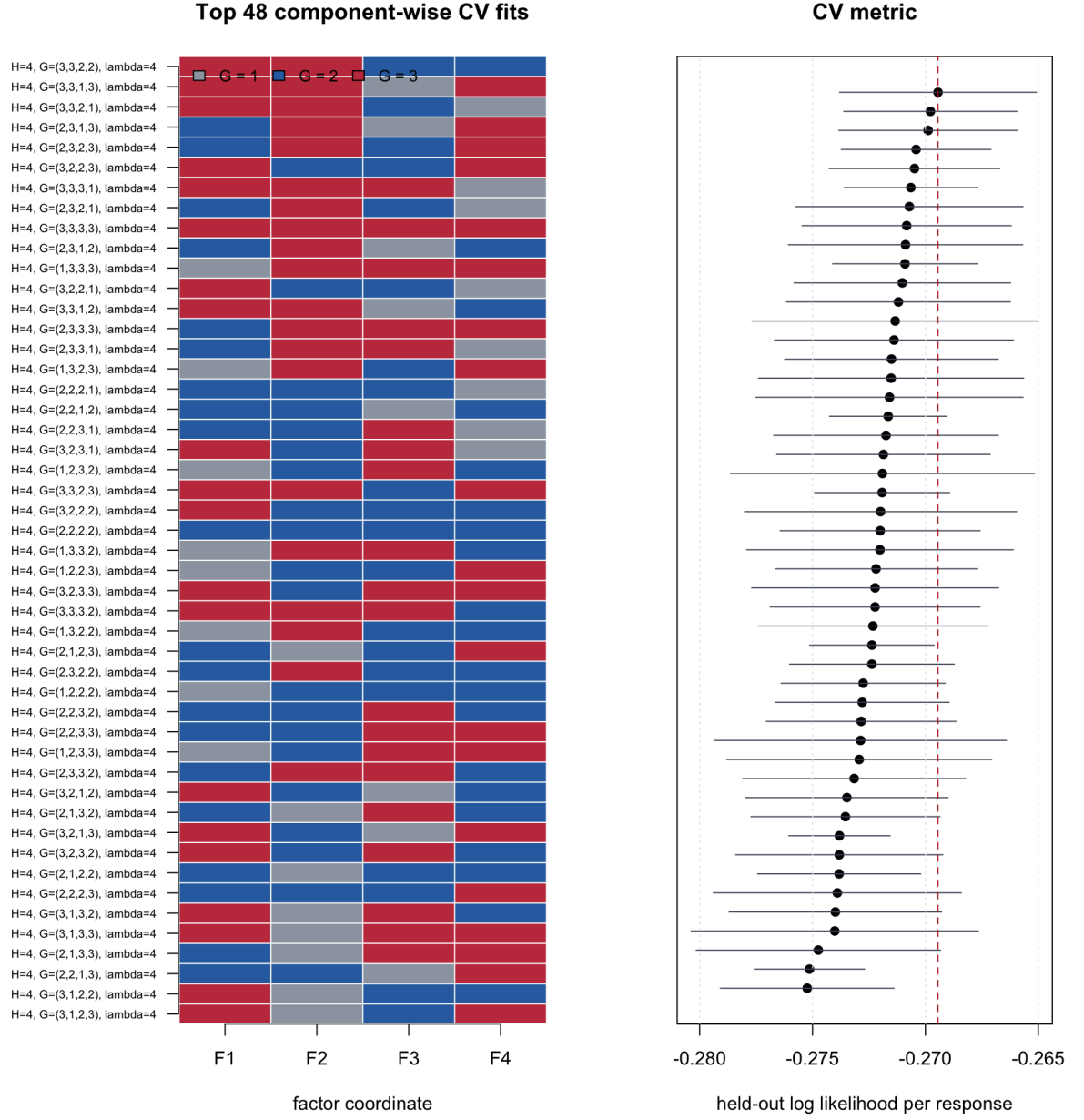


Figure 1: Component-wise CV metrics for $H=4$, $\lambda=4$

This is useful substantively: F3 is retained as a continuous axis, while F1, F2, and F4 are allowed to form discrete model subtypes.

Fitted Mixture Structure

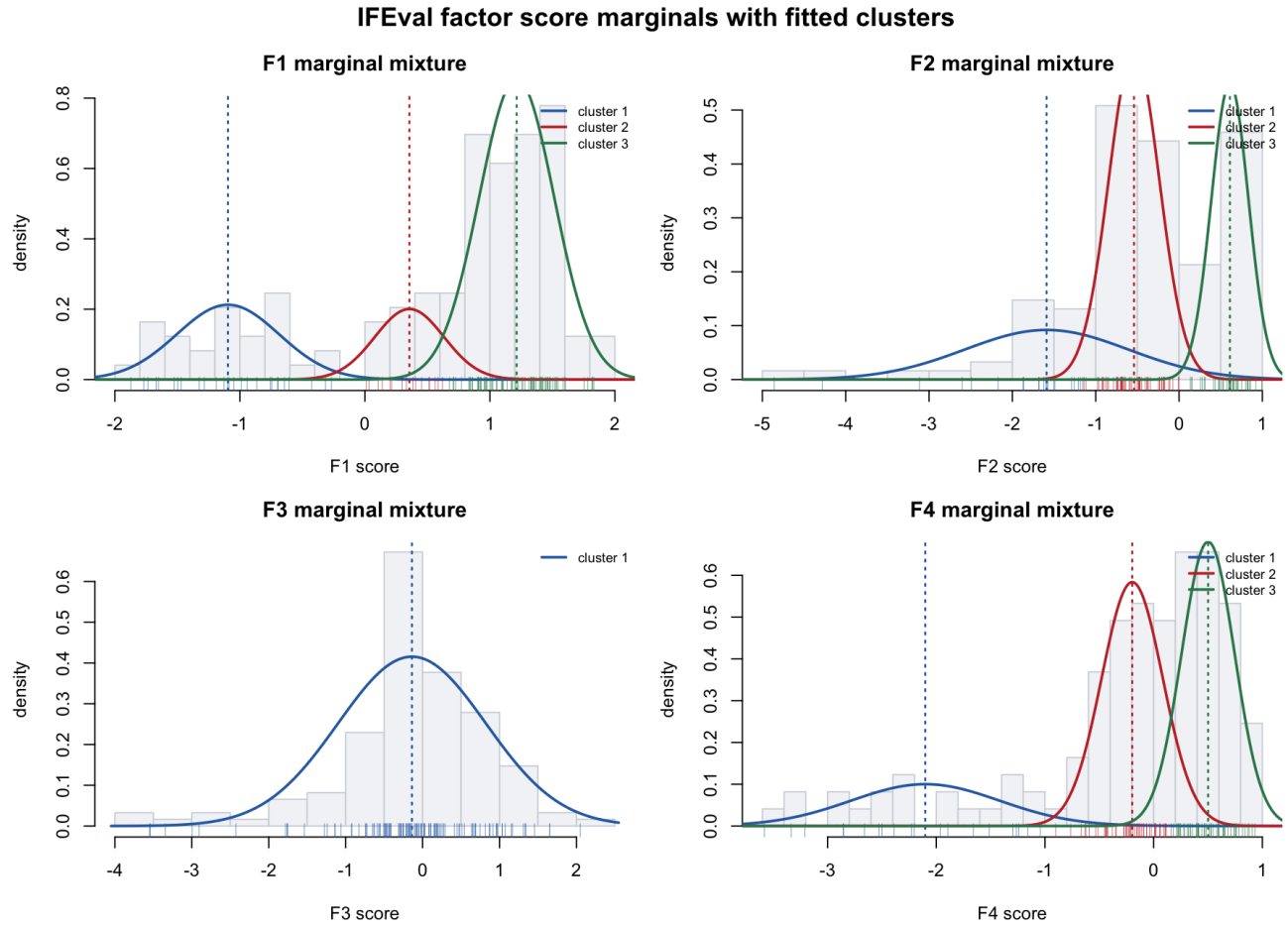


Figure 2: Marginal mixture fits

The selected fit uses component counts $\mathbf{G}=(3,3,1,3)$. Thus F1, F2, and F4 have discrete low/middle/high model groups, while F3 is deliberately modeled as one continuous Gaussian coordinate. The group labels below are interpreted using the reported factor scores and MAP group assignments in `openeval_model_factor_scores_profiles.csv`.

Factor	Groups	Score interpretation	Substantive interpretation
F1	1/2/3: 27 / 16 / 79 models	low / middle / high	Broad instruction-following ability.
F2	1/2/3: 21 / 61 / 40 models	low / middle / high	Language, case, and script-control ability.
F3	1: 122 models	continuous	Creative lexical-control ability, not a discrete subtype.
F4	1/2/3: 20 / 50 / 52 models	low / middle / high	Structured-output and local-formatting ability.

Empirically, the MAP groups have the following reported factor-score summaries. The weights in this table are empirical occupancies, so they are directly tied to the plotted model profiles.

Factor	Group	Weight	Mean score	SD score
F1	low	0.221	-1.085	0.446
F1	middle	0.131	0.336	0.193
F1	high	0.648	1.221	0.287
F2	low	0.172	-2.004	1.164
F2	middle	0.500	-0.545	0.346
F2	high	0.328	0.593	0.237
F3	continuous	1.000	-0.138	0.960
F4	low	0.164	-2.169	0.770
F4	middle	0.410	-0.235	0.284
F4	high	0.426	0.516	0.231

The fitted mixture is therefore not just a ranking model. It separates broad IFEval strength from more specific forms of constraint-following. For example, a model can be in the high-F1 group while only middle on F2 or F4, meaning it generally follows instructions well but is less reliably specialized for language/case/script constraints or structured-output constraints.

LLM Scores and Mixture Profiles

The next two plots show the fitted LLM coordinates. Models are ordered from left to right by empirical IFEval accuracy.

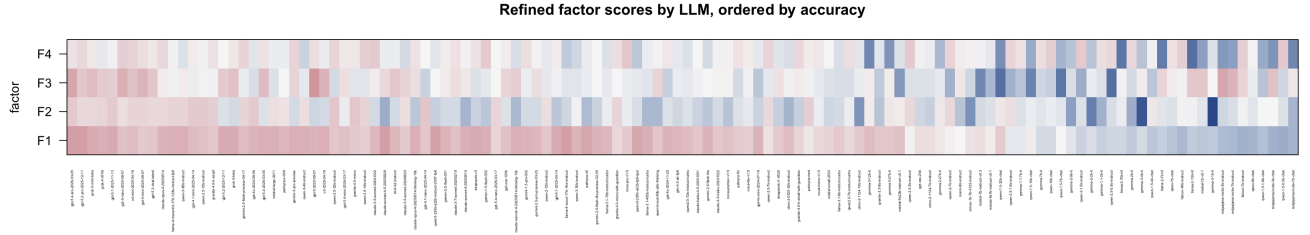


Figure 3: Refined factor scores by LLM

The continuous score heatmap shows the broad role of F1: the highest-accuracy models tend to have high F1 scores, while weaker models tend to move downward on F1. F2, F3, and F4 are less monotone in overall accuracy. They capture different styles of constraint-following rather than just “better versus worse” performance.

The group heatmap translates the continuous scores into factor-wise MAP mixture assignments. F3 is intentionally a single Gaussian coordinate, so every model has group 1 on F3. F1, F2, and F4 show discrete heterogeneity: models can be strong on the broad instruction-following axis while belonging to different F2 or F4 subtypes.

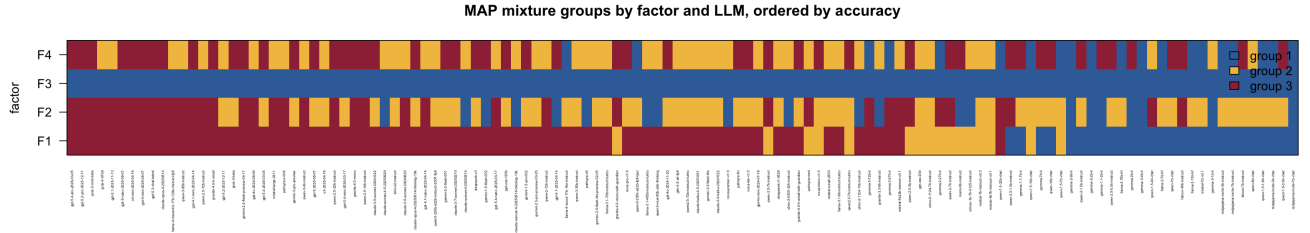


Figure 4: MAP mixture groups by LLM

LLM Ability Profiles

The marginal factor coordinates give a more nuanced model profile than overall accuracy alone. The joint profiles are interpreted in detail after the loading matrix because the meaning of each coordinate is clearer once the item loadings are inspected.

- **F1: broad instruction-following ability.** High F1 models include `claude-sonnet-4-5-20250929`, `llama-4-scout-17b-16e-instruct`, `gpt-5.4-pro-2026-03-05`, and `gpt-5.2-pro-2025-12-11`. Low F1 models include small or older chat/instruct models such as `qwen-1.5-0.5b-chat`, `falcon-7b-instruct`, and `redpajama-incite` variants. This is the closest coordinate to a broad IFEval ability axis.
- **F2: language, case, and script-control ability.** High F2 models include `qwen-2.5-32b-instruct`, `granite-4.0-micro`, `gpt-4.1-nano-2025-04-14`, `gpt-5.1-2025-11-13`, and `gpt-4o-2024-08-06`. Low F2 models include `gemma-3-1b-it`, `gemma-3-4b-it`, and `gemma-2-2b-it`. This axis separates models on lowercasing, single-language output, and script-control constraints.
- **F3: continuous creative lexical-control ability.** High F3 models include `gpt-5-2025-08-07`, `gpt-5.4-pro-2026-03-05`, `gpt-5-nano-2025-08-07`, `gpt-5-mini-2025-08-07`, and `o3-2025-04-16`. Low F3 models include several Qwen and Mistral-family entries such as `qwen-1.5-32b-chat`, `qwen-2.5-0.5b-instruct`, and `mistral-7b-instruct-v0.3`. Because F3 is modeled with one Gaussian component, these are high/low positions on a continuous axis rather than discrete subtypes.
- **F4: structured-output and local-formatting ability.** High F4 models include `nova-pro-v1:0`, `claude-3-5-sonnet-20241022`, `gemini-1.5-flash-002`, `qwen-2.5-14b-instruct`, and `gpt-5-nano-2025-08-07`. Low F4 models include llama-2 base models, `gemma-3-12b-it`, `redpajama-incite-7b-chat`, and `qwen-1.5-32b-chat`. This coordinate appears to capture JSON wrapping, exact endings, sentence-count limits, and similar formatting demands.

Loading Matrix

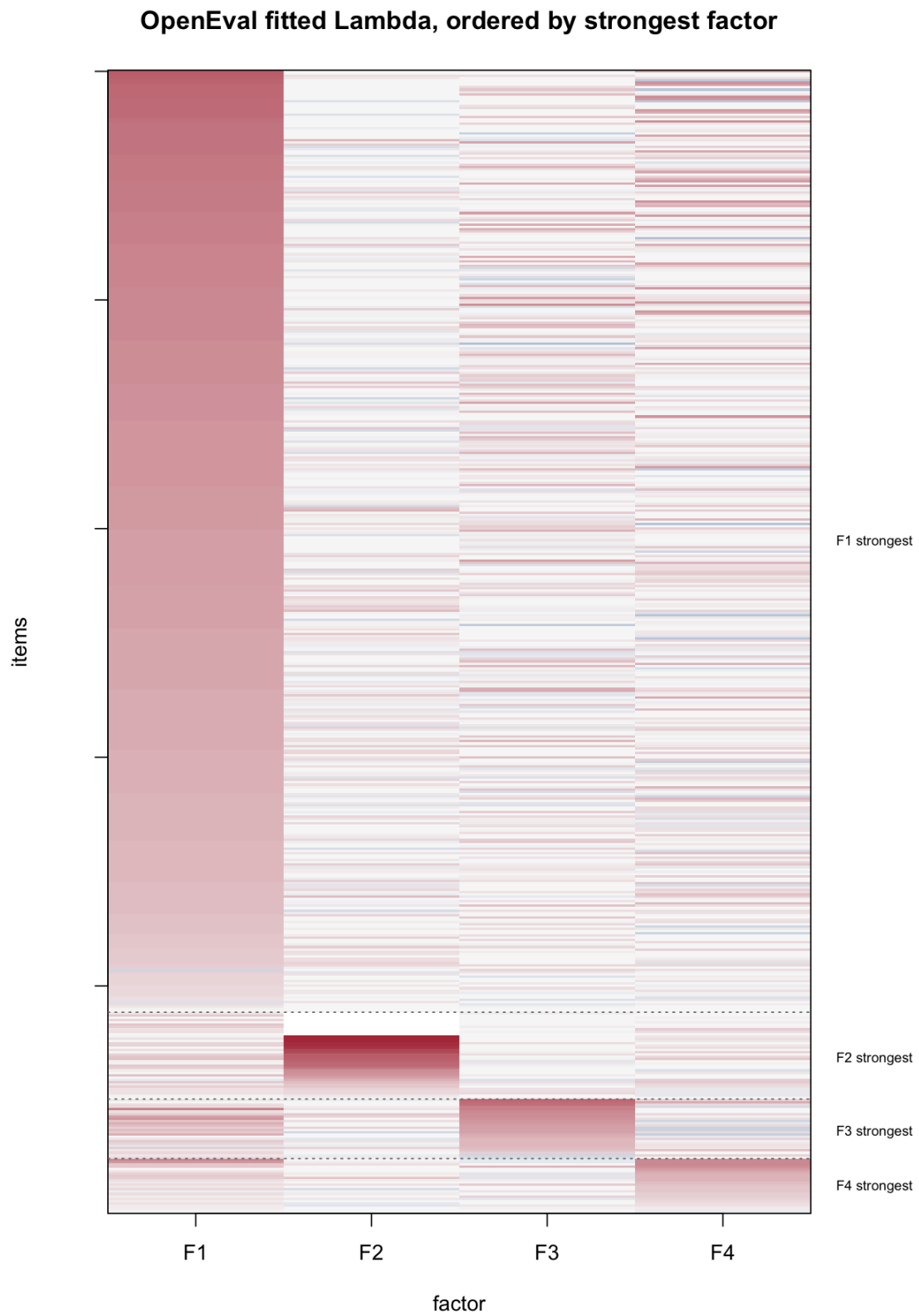


Figure 5: Lambda ordered by strongest factor

The heatmap displays the fitted item loading matrix, Λ , after permuting the 500 item rows. Each row is one IFEval item and each column is one factor. Red entries are positive loadings, blue entries are negative loadings, and white entries are near zero. Larger absolute values mean that the item’s success probability changes more strongly with that factor score.

The row ordering is diagnostic, not part of the fitted model. For each item j , define

```
strongest_factor_j = argmax_h |lambda_jh|
max_abs_loading_j = max_h |lambda_jh|.
```

Rows are sorted by `strongest_factor_j`, then by decreasing `max_abs_loading_j`. Thus, the apparent blocks are not imposed by the model; they are a visualization of the learned loading pattern. The large upper block consists of items whose strongest absolute loading is on F1. The smaller lower blocks consist of items whose strongest loading is on F2, F3, or F4. The fact that these blocks become visible after a genuine row permutation is evidence that the fitted loadings have a broad-plus-specific structure: one dominant general IFEval factor and several smaller private axes.

Among active items with $|\text{loading}| > 0.1$, the strongest-factor counts are:

Strongest factor	Active items
F1	410
F2	37
F3	26
F4	23

The numeric loading summary also shows this structure:

Factor	Mean abs loading	$ \lambda > 0.5$	$ \lambda > 1.0$
F1	0.863	396	199
F2	0.240	38	28
F3	0.203	63	13
F4	0.222	69	18

Factor Interpretation

The loadings give the item-level interpretation of the factors. A strong solo-loading item has one dominant loading and small loadings on the other coordinates, so it is useful for naming a factor. A cross-loading item has large loadings on more than one coordinate, so it reveals tasks that require a combination of abilities.

Strong solo-loading examples include:

Primary factor	Item	Loadings (F1,F2,F3,F4)	Example constraint
F1	ifeval_20260421T021146Z_242758	0.863, 0.000, 0.000, 0.000)	Repeat the request exactly, then answer with a title in double angle brackets.
F2	ifeval_20260421T021146Z_293137	0.000, 3.071, 0.105, 0.000)	Product description entirely in lowercase.
F3	ifeval_20260421T021146Z_207167	-0.087, 1.637, 0.343)	Write a rubric as a poem while using the letter w fewer than two times.
F4	ifeval_20260421T021146Z_369479	0.248, 0.000, 1.207)	Use fewer than 10 sentences and end with a required sentence.

F1: Broad Instruction Following

F1 is the dominant general axis. High-loading items often combine ordinary task completion with explicit constraints, such as repeating the request exactly, ending with a required phrase, using all caps, or wrapping text in prescribed delimiters.

Representative high-F1 items include:

Item	Loading	Example constraint
ifeval_20260421T021146Z_127	1.785	End the response with a specified sentence and no words after it.
ifeval_20260421T021146Z_242	1.758	Repeat the request exactly, then answer with a title in double angle brackets.
ifeval_20260421T021146Z_362	1.742	Repeat the request word-for-word before answering.
ifeval_20260421T021146Z_6	1.701	Write two labeled all-caps paragraphs.

F2: Language, Case, and Script Control

F2 is more specialized. The strongest positive F2 items emphasize lowercasing, single-language output, and constraints on allowed alphabets or scripts.

Representative high-F2 items include:

Item	Loading	Example constraint
ifeval_20260421T021146Z_293	3.071	Product description entirely in lowercase.
ifeval_20260421T021146Z_409	2.967	English-only response, all lowercase.
ifeval_20260421T021146Z_280	2.963	Hindi poem, only Hindi, no other language.
ifeval_20260421T021146Z_292	2.822	Haiku in lowercase with no capital letters.

F3: Creative Lexical Constraint Axis

F3 is best treated as continuous in this fit. Its high-loading items often require creative generation under severe lexical constraints, such as avoiding or limiting a common letter while still producing a poem, song, quiz, or long essay.

Representative high-F3 items include:

Item	Loading	Example constraint
ifeval_20260421T021146Z_277	1.637	Write a rubric as a poem while using the letter w fewer than two times.
ifeval_20260421T021146Z_477	1.565	Write a song while using the letter a at most once.
ifeval_20260421T021146Z_23	1.516	Write a logic quiz while using the letter t at most once.
ifeval_20260421T021146Z_147	1.324	Write a long essay of at least 50 sentences.

The Gaussian choice for F3 is important: it suggests this ability varies continuously across LLMs rather than forming a stable discrete subtype in the current data.

F4: Structured Output and Local Formatting

F4 captures a smaller but interpretable formatting axis. High-loading items frequently require JSON wrapping, sentence-count limits, required endings, highlighted sections, or other local structural constraints.

Representative high-F4 items include:

Item	Loading	Example constraint
ifeval_20260421T021146Z_269	1.219	Wrap the entire response in JSON format.
ifeval_20260421T021146Z_58	1.213	Answer in JSON format, with markdown ticks allowed.
ifeval_20260421T021146Z_359	1.207	Use fewer than 10 sentences and end with a required sentence.
ifeval_20260421T021146Z_33	1.173	Use a repeated keyword, fewer than six sentences, and highlighted text sections.

Cross-Loading Items

Cross-loading items reveal tasks that mix multiple skill demands. Using the rule that two factors both have $|\text{loading}| \geq 0.5$, the observed pair types are F1/F2, F1/F3, F1/F4, and F3/F4. No F2/F3 or F2/F4 items met this threshold in this fit.

- **F1/F2:** ifeval_20260421T021146Z_245, with loadings (F1,F2,F3,F4) = (1.001, 0.707, 0.000, 0.416). The item asks for exactly three names using markdown bullet points, combining broad instruction following with list-format control.
- **F1/F3:** ifeval_20260421T021146Z_80, with loadings (1.300, 0.198, 1.301, -0.018). The item asks for a 30-line poem with exact sentence and punctuation constraints, mixing broad instruction following with creative lexical/form control.
- **F1/F4:** ifeval_20260421T021146Z_269, with loadings (1.199, 0.000, -0.177, 1.219). The item requires JSON wrapping plus ordinary semantic interpretation, mixing general instruction following with structured-output formatting.
- **F3/F4:** ifeval_20260421T021146Z_477, with loadings (0.000, 0.000, 1.565, 0.816). The item asks for a song while using the letter **a** at most once, combining creative generation with strict local lexical control.

The cross-loading pattern reinforces the heatmap interpretation. The learned matrix is not pure block diagonal. It is closer to a broad general factor with smaller specialized factors that partially overlap when an item combines content, language/case, lexical creativity, and output-format constraints.

Joint LLM Ability Profiles

The joint MAP profile has the form F1-F2-F3-F4. Since F3 has one Gaussian component, the third entry is always 1; the continuous F3 score still matters, but it does not create a discrete cluster. For F1, F2, and F4, group 3 is the high-score group, group 2 is the middle group, and group 1 is the low-score group.

- **Profile 3-3-1-3: high broad ability, high language/case/script control, continuous F3, high structured-output control.** This is the strongest joint profile: 14 models, mean IFEval accuracy 0.948, range 0.910 to 0.992. Examples include `gpt-5.4-pro-2026-03-05`, `gpt-5.2-pro-2025-12-11`, `grok-3-mini-beta`, and `gpt-5-nano-2025-08-07`. These models excel broadly and also retain strength on specialized case/language and formatting constraints.

- **Profile 3-3-1-2: high F1 and F2 but only middle F4.** This profile has 14 models with mean accuracy 0.919. Examples include `grok-4-0709`, `gpt-5.1-2025-11-13`, `llama-4-maverick-17b-128e-instruct-fp8`, and `qwen-3-80b-instruct`. These models are strong overall and strong on language/case/script tasks, but their relative weakness is the local structured-output axis: JSON wrapping, exact endings, sentence-count limits, and similar formatting constraints.
- **Profile 3-2-1-3: high F1 and F4 but middle F2.** This profile has 17 models with mean accuracy 0.895. Examples include `grok-3-beta`, `gpt-5.4-2026-03-05`, `gemini-3-pro-preview`, and `gpt-5-2025-08-07`. These models look strong on broad instruction-following and structured output, but they are less specialized for all-lowercase, single-language, or script-restricted tasks.
- **Profile 3-1-1-2: high F1 but low F2 and middle F4.** This profile has 10 models with mean accuracy 0.865. Examples include `claude-sonnet-4-5-20250929`, `claude-sonnet-4-20250514`, `claude-sonnet-4-20250514-thinking-10k`, and `palmyra-x5`. The interpretation is not that these models are weak overall; rather, they sit high on the broad general axis but suffer specifically on language/case/script control and are only middle on local formatting.
- **Low-F1 profiles identify broad instruction-following failures.** Profiles such as 1-2-1-1 and 1-1-1-1 contain models with much lower overall accuracy. Examples include `llama-2` base models, `mistral-7b-v0.1`, `gemma-2-2b-it`, `gemma-3-4b-it`, and `redpajama-incite-7b-chat`. Some low-F1 models can still be assigned to a high F4 group, but this does not compensate for broad failures on the dominant F1 axis.

This joint-profile view is the main practical advantage of the fitted mixture. Overall accuracy says which models are better on average, while the profile identifies how they are better or worse: broad instruction-following, language/case/script control, creative lexical control, or structured-output formatting.

Working Interpretation

The $G=(3,3,1,3)$ fit gives a clean compromise between predictive performance and interpretability. F1 is the broad general IFEval ability. F2 and F4 describe discrete model subtypes for specialized constraint-following modes. F3 remains a continuous creative-control axis, especially for generative tasks with severe lexical restrictions.

This structure is more interpretable than forcing F3 to have multiple mixture components: when F3 is given two or three components, the extra components tend to model central concentration rather than a meaningful left-tail or high-skill subtype.